Week 4 Reference Sheet: Math Operations & Operators

CSC-125-201 | Use this as a reference when completing Homework 4

Homework Topics

- Arithmetic operators: `+` `-` `*` `/` `//` `%` `**`
- Floor division and modulo uses
- Operator precedence (PEMDAS)
- Comparison operators and booleans
- Assignment operators: `+=` `-=` `*=` `/=`

Quick Reference

Arithmetic Operators

Operator	Name	Example	Result
`+`	Addition	`17 + 5`	22
`-`	Subtraction	`17 - 5`	12
`*`	Multiplication	`17 * 5`	85
`/`	Division (float)	`17 / 5`	3.4
`//`	Floor Division	`17 // 5`	3
`%`	Modulo (remainder)	`17 % 5`	2
`**`	Exponentiation	`2 ** 5`	32

Floor Division & Modulo in Practice

```
# Division always returns a float; // rounds down; % gives remainder
10 / 3     # 3.3333...
10 // 3    # 3
10 % 3     # 1     (relationship: 10 = 3*3 + 1)

# Time conversion
total_min = 135
hours = total_min // 60    # 2
mins = total_min % 60      # 15

# Even/odd check
number % 2 == 0    # True if even, False if odd
```

Operator Precedence (PEMDAS)

Priority	Operator(s)	Description
1 (highest)	`()`	Parentheses
2	`**`	Exponentiation
3	`*` `/` `//` `%`	Multiply, Divide, Floor, Mod
4	`+` `-`	Add, Subtract
5	`==` `!=` `<` `>` `<=` `>=`	Comparisons
6 (lowest)	`=` `+=` `-=` etc.	Assignment

```
2 + 3 * 4                      # 14 (not 20)
(2 + 3) * 4                    # 20
average = (s1 + s2 + s3) / 3   # parentheses for clarity
```

Comparison Operators

Compare two values → returns `True` or `False`

Operator	Meaning	Example (`age = 20`)	Result
`==`	Equal to	`age == 18`	False
`!=`	Not equal	`age != 18`	True
`<`	Less than	`age < 65`	True
`>`	Greater than	`age > 18`	True
`<=`	Less/equal	`age <= 21`	True
`>=`	Greater/equal	`age >= 18`	True

Boolean Expressions

```
age = 20
score = 85

can_vote = age >= 18          # True
is_passing = score >= 70      # True
is_perfect = score == 100     # False
is_teen = 13 <= age <= 19     # False (chained comparison)
```

Assignment Operators

Shorthand	Equivalent	Example (start: `x = 10`)	Result
`x += 5`	`x = x + 5`	`10 + 5`	15
`x -= 3`	`x = x - 3`	`15 - 3`	12
`x *= 2`	`x = x * 2`	`12 * 2`	24
`x /= 4`	`x = x / 4`	`24 / 4`	6.0

Common Errors

Error	Cause	Fix
Wrong result	Missing parentheses	Use `()` to control precedence
`TypeError`	Math with strings from `input()`	Convert with `int()` or `float()`
Confusion	`=` vs `==`	`=` assigns, `==` compares
`ZeroDivisionError`	Dividing by zero	Check divisor before dividing

Additional Resources

- Corey Schafer - Python Operators (video)
- W3Schools - Python Operators
- LearnPython.org - Basic Operators (interactive)
- Real Python - Operators and Expressions
- W3Schools - Python Booleans